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SYNOPSIS ON

“AI-Powered Real-Time Urban Traffic Congestion Prediction Using Distributed Big Data Processing”

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Department of Computer Science & Engineering

2024-2025

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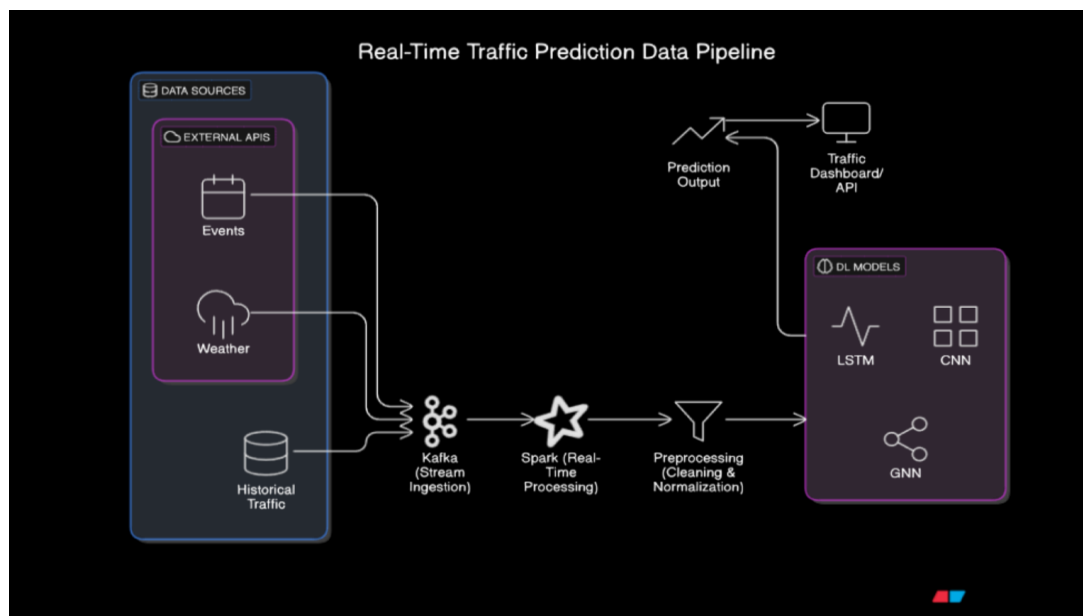
Project Batch Number: 01

Abstract

In the evolving landscape of smart cities, managing urban traffic effectively has become a significant challenge due to diverse data sources and complex patterns of vehicular movement. This project proposes an integrated prediction model that leverages some particular deep learning architecture to analyse city-wide traffic conditions. By incorporating data such as vehicle speed, estimated time of arrival (ETA), weather conditions, and road structure from multiple heterogeneous sources, the system creates a hybrid feature space that enhances congestion forecasting accuracy.

Introduction

Smart cities aim to enhance the quality of life by integrating advanced technologies into urban infrastructure. Among the many challenges faced in this domain, traffic congestion remains a primary concern. Rapid urbanization and increasing vehicular load strain existing road networks, leading to extended travel times, environmental degradation, and economic loss. Traditional prediction models often fall short due to their limited scope and inability to incorporate diverse data types. To address this, our project introduces a novel predictive framework that utilizes data from various sources — including GPS devices, weather APIs, and mapping services — processed through a hybrid deep learning approach. This method enables accurate and timely forecasting of traffic congestion, contributing to more efficient urban mobility solutions.



Problem Statement

Urban areas are experiencing heightened traffic congestion, which leads to significant delays, resource wastage, and economic drawbacks. Predicting traffic patterns is complicated by the dynamic and multifaceted nature of contributing factors such as weather, road conditions, time of day, and vehicular flow. Existing models often lack the capability to integrate and process data from multiple heterogeneous sources effectively, resulting in sub-optimal prediction outcomes. There is a pressing need for a robust, scalable, and intelligent traffic forecasting system that can accommodate the spatial-temporal complexities of urban transport networks.

Objectives

1. To develop a traffic prediction model that accurately forecasts congestion levels on city roads.
2. To integrate heterogeneous data sources including speed data, ETA, weather, and road network characteristics
3. To design and implement a deep learning framework for enhanced spatial-temporal traffic analysis.
4. To evaluate the performance of the proposed model against traditional machine learning methods.
5. To provide actionable insights for traffic management systems in smart city infrastructure.

LITERATURE SURVEY

Title	Authors	Published Year	Methodology	Limitations
Traffic Prediction in Smart Cities Based on Hybrid Feature Space	N. Zafar et al.	2022	Hybrid GRU-LSTM model using heterogeneous data sources	Limited to Islamabad city data; high preprocessing complexity
A Comprehensive Evaluation of Deep Learning-Based Techniques for Traffic Prediction	J. Mena-Oreja and J. Gozalvez	2020	Survey of deep learning models including CNNs, RNNs, LSTM, GRU	Lacks real-world deployment examples
AI-Driven Traffic Congestion Management	S. P. S. Rathore et al.	2025	Predictive analytics with AI-based congestion models	Focuses more on conceptual framework, lacks model specifics
AI-Powered Traffic Management	M. D. Malathi et al.	2025	AI for congestion detection and signal regulation using ML	Limited dataset scalability
An Automated Hyperparameter Search-Based Deep Learning Model	H. Yi and K.-H. N. Bui	2021	DL model with automated hyperparameter tuning	High computational cost
Enhancing Traffic Flow with IoT-based Intelligent Traffic System	M. H. Kabir and M. S. Islam	2024	IoT integration with intelligent traffic management	Security and integration challenges
Entropy-Based Traffic Flow Labelling	M. Z. Mehdi et al.	2022	CNN model using entropy-based feature labeling IoT-enabled traffic signal manipulation for emergency	Doesn't consider dynamic road conditions
IoT-Based Traffic Control System for Emergency Vehicles	K. P. Kavitha et al.	2024	IoT-enabled traffic signal manipulation for emergency	Limited to emergency vehicle focus
LSTM Model for Traffic Congestion Prediction	Y. Chen et al.	2016	LSTM model with open traffic data	Older data and limited to LSTM

MF-TCPV: A ML and Fuzzy Evaluation Framework	L. Li et al.	2020	ML with fuzzy logic for traffic visualization	Complexity in tuning fuzzy parameters
RouteRover: AI-Enabled Traffic Prediction	N. S et al.	2024	AI for congestion prediction and route optimization	Tailored for Indian urban traffic
Traffic Management Using Machine Learning	S. Telsang et al.	2024	ML models for signal control and traffic flow	Focused on signal manipulation only
Traffic Congestion Warning Based on Data Modelling	C. Zhong et al.	2023	Predictive modeling for congestion warnings	Simplistic modeling assumptions
Simulation-Based Traffic Management	S. Satkunarajah and R. Puvanendran	2023	Simulation-based traffic flow reduction model	Requires real-time data for effectiveness
A Hybrid Method of Traffic Congestion Prediction	T. Zhang et al.	2023	Hybrid deep learning and control strategy	Implementation challenges in dynamic systems
Highway Traffic Flow Control with Fuzzy Control	B. Nie and B. Ma	2023	Fuzzy logic model for highway traffic	Limited real-time adaptability
Real-Time Adaptive Traffic Control	S. Shankaran and L. Rajendran	2021	Adaptive traffic signal using ML	Scalability to large networks untested
Scalable Big Data for Intelligent Traffic Control	Y. Zheng	2024	Big data analytics for traffic control	Requires substantial infrastructure and investment

System Design Architecture

The system architecture is structured around a multi-stage data processing and predictive modelling pipeline:

1. **Data Collection:** Aggregation of traffic-related data from multiple sources Google Maps API for ETA, Open Street Maps (OSM) for road topology, and weather data from third-party APIs.
2. **Data Preprocessing:** Includes map matching, cleaning, zero-speed adjustments, and spatial-temporal normalization to ensure consistent format and accuracy.
3. **Hybrid Feature Space Formation:** All cleaned data streams are merged into a unified schema that captures both spatial and temporal attributes.
4. **Predictive Modelling:** Deep Learning models processes the hybrid data to learn from both recent and long-term trends, improving congestion prediction.
5. **Output Generation:** The system classifies traffic conditions into categories such as smooth, congested, and highly congested and provides insights for urban planners and traffic control systems.

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